24 种锂基材料的 Bader 电荷、ELF 分布与 Li 空位形成能计算报告

GPU-VASP 6.5.1 统一参数计算与跨材料比较

项目	内容
材料范围	24 种锂基材料；46 个对称不等价 Li 空位
电子结构结果	24 套 bulk Bader 电荷与 ELF 体数据
统一关键参数	ENCUT = 450 eV；EDIFFG = −0.05 eV Å^{-1}
结构图渲染	CrystalStudio 0.12.30 后台模式；a/b/c 均为−0.01–1.01；黑色线框；线宽 1.0
报告日期	2026-09-20

本报告仅依据本项目实际 VASP 输入/输出、Bader 输出、ELFCAR 和结构文件形成结论。所有能量、收敛状态和计算参数均由本地交付文件可追溯复核。

摘要

本项目完成 24 种锂基材料的体相结构弛豫、静态电子结构、Bader 电荷和电子局域函数（ELF）计算，并在尽量接近的 Li 空位浓度建立 46 个对称不等价中性 Li 空位模型。全部 24 个 bulk 弛豫、24 个 bulk 静态、46 个缺陷弛豫、46 个缺陷静态及 bcc Li 参考相均成功结束；70 个结构弛豫均达到力收敛，最终最大离子力不超过 0.04985 eV Å⁻¹。

以同参数 bcc Li 每原子能量 μLi = −1.885472 eV 为化学势，按 Ef = Edefect − Ebulk + μLi 计算中性 Li 空位形成能。24 种材料的最低位点形成能介于 0.241–6.667 eV：LiC6 最低（0.241 eV），Li3N、Li6CoO4、Li6MnO4、Li3P 和 Li2MoO3 也低于 2.0 eV；LiF 最高（6.667 eV）。全部位点形成能均为正，说明在 Li 金属参考化学势和本计算设置下，自发生成中性 Li 空位在 0 K 总能层面均不利。

Bader 分析显示 Li 平均失电子量为+0.507–+0.920 e。最低形成能与 Li 平均 Bader 失电子量、空位浓度和全局平均 ELF 的 Pearson 相关系数分别为 0.077、0.124 和−0.177，均未显示强线性关系；这表明跨不同化学体系时，Li 空位形成能不能由单一电荷或单一 ELF 统计量充分解释。ELF≥0.8 体积分数与形成能的 Spearman 相关为 ρ=0.453（p=0.026），可作为探索性趋势，但受到材料化学组成、晶体结构与缺陷局域弛豫共同影响，不应作因果解释。

1. 计算范围与完成情况

任务类型	计划	完成	核验结果
bulk 弛豫	24	24	全部达到 EDIFFG 阈值
bulk 静态	24	24	全部含 CHGCAR、AECCAR0/2、ELFCAR、ACF.dat
Li 空位弛豫	46	46	全部达到 EDIFFG 阈值
Li 空位静态	46	46	全部正常结束
bcc Li 参考	1 套	1 套	同参数化学势

所有生产作业均在 GPU 分区运行，并调用 /public/software/vasp/GPU/vasp.6.5.1/vasp_std。统计共 140 个材料计算任务（不含 Li 参考相），FAILED 标志为 0。bulk 和缺陷静态任务均在各自弛豫结束后执行，依赖链完整。

2. 结构模型与可比性设计

初始结构取自 Materials Project 对应条目，随后以原胞/常规胞为基础构建尽量小的超胞，使 24 种材料的 Li 空位浓度集中在 2.50%–6.25%。该策略避免了对低 Li 含量材料使用过大的统一超胞，同时确保同一材料内所有不等价 Li 位点使用相同超胞、相同 k 点网格与相同电子结构参数。跨材料仍存在 2.50%–6.25%的有限浓度差异，因此形成能排序应理解为在接近而非完全相同缺陷浓度下的比较。

序号	材料	MP-ID	超胞	原子数	Li 数	VLi/%	位点数	k 网格
1	Li5FeO4	mp-19511	1×1×1	80	40	2.50	5	2×2×2
2	Li6CoO4	mp-18925	2×1×1	44	24	4.17	2	2×2×2
3	Li2NiO2	mp-19308	2×2×2	40	16	6.25	1	2×2×2
4	Li2CuO2	mp-4711	2×2×2	40	16	6.25	1	3×2×2
5	Li4SiO4	mp-11737	2×1×1	36	16	6.25	4	2×3×2
6	Li6MnO4	mp-770533	2×1×1	44	24	4.17	2	2×2×2
7	Li6ZnO4	mp-14495	2×1×1	44	24	4.17	2	2×2×2
8	Li5AlO4	mp-15960	1×1×1	80	40	2.50	5	2×2×2
9	Li2MoO3	mp-694855	2×2×1	48	16	6.25	3	2×2×3
10	Li8ZrO6	mp-755225	2×1×1	30	16	6.25	2	2×3×2
11	Li3BO3	mp-27275	2×1×1	56	24	4.17	3	2×2×2
12	LiNO3	mp-8180	2×2×2	80	16	6.25	1	2×2×2

序号	材料	MP-ID	超胞	原子数	Li 数	VLi/%	位点数	k 网格
13	Li2O2	mp-841	2×2×1	32	16	6.25	2	2×2×2
14	Li2O	mp-1960	2×2×2	24	16	6.25	1	2×2×2
15	Li2S	mp-1153	2×2×2	24	16	6.25	1	2×2×2
16	Li3N	mp-2251	2×3×1	24	18	5.56	2	2×2×3
17	Li3P	mp-736	1×3×1	24	18	5.56	2	3×1×2
18	Li2Se	mp-2286	2×2×2	24	16	6.25	1	2×2×2
19	LiF	mp-1138	2×2×4	32	16	6.25	1	3×3×2
20	LiC6	mp-1001581	4×2×2	112	16	6.25	1	1×2×2
21	LiC	mp-1378	2×2×2	32	16	6.25	1	2×2×2
22	Li2CO3	mp-3054	2×2×1	48	16	6.25	1	2×2×2
23	LiCO2	mp-1190287	2×2×1	64	16	6.25	1	2×2×2
24	Li(CO)2	mp-546920	2×2×2	80	16	6.25	1	2×1×1

3. 计算方法与参数证据

采用 VASP 进行自旋极化平面波密度泛函理论计算，[1] 交换关联泛函为 PBE 广义梯度近似，[2] 离子—电子相互作用采用 PAW 方法。[3,4]

所有计算均使用 ENCUT = 450 eV、EDIFF = 1×10^{-5} eV、PREC = Accurate、ISMEAR = 0、SIGMA = 0.05 eV、LREAL = .FALSE.、LASPH = .TRUE.。几何优化采用 EDIFFG = −0.05 eV $Å^{-1}$。静态任务采用 IBRION = −1 和 NSW = 0。

bulk 静态计算输出 CHGCAR、AECCAR0、AECCAR2 与 ELFCAR。Bader 分析采用价电子密度并以 AECCAR0+AECCAR2 作为全电子参考，[5] 原子净失电子量定义为 q = Zval − QBader，q > 0 表示失电子。ELF 按 Becke–Edgecombe 定义解释。[6]

ELF 二维图使用每个结构沿 c 方向标准差最大的代表性网格切片，所有材料使用统一 0–1 颜色范围；全局平均值与 ELF≥0.8 体积分数由完整三维 ELFCAR 网格统计。该切片策略用于展示局域化空间差异，不等同于指定晶面上的严格二维积分。

参数/项目	实际值	适用范围	证据
Program	vasp.6.5.1 10Mar25 (build Nov 26 2025 15:47:16) complex	all VASP tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Executable	/public/software/vasp/GPU/vasp.6.5.1/vasp_std	all VASP tasks	/public/home/pai/9yue/run_gpu.sbatch
Partition/GPU	GPU partition; 1 GPU	all VASP tasks	/public/home/pai/9yue/run_gpu.sbatch
ENCUT	450 eV	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
EDIFF	1E-5 eV	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
EDIFFG	-0.05 eV Å^-1	relaxation tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_relax\OUTCAR
IBRION	2 initially; selected long-running continuations used 1	relaxation tasks	individual OUTCAR files
Static IBRION/NSW	IBRION=-1; NSW=0	all final static tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Exchange-correlation	GGA-PBE	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
PAW	PAW_PBE datasets; species-specific titles retained in OUTCAR	all tasks	individual OUTCAR files
Spin	ISPIN=2	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Smearing	ISMEAR=0; SIGMA=0.05 eV	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Precision/projectors	PREC=Accurate; LREAL=.FALSE.; LASPH=.TRUE.	all tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Bulk static outputs	LCHARG=.TRUE.; LAECHG=.TRUE.; LELF=.TRUE.	24 bulk static tasks	outputs\计算原始数据\DFT 核心输入输出\mp-19511\bulk_static\OUTCAR
Bader definition	q = Z_val - Q_Bader; q>0 denotes electron loss	bulk Bader analysis	ACF.dat and OUTCAR ZVAL
Li chemical potential	-1.88547189 eV atom^-1	neutral Li-vacancy formation energies	outputs\计算原始数据\DFT 核心输入输出\Li_bcc_reference\OUTCAR

4. Li 空位形成能

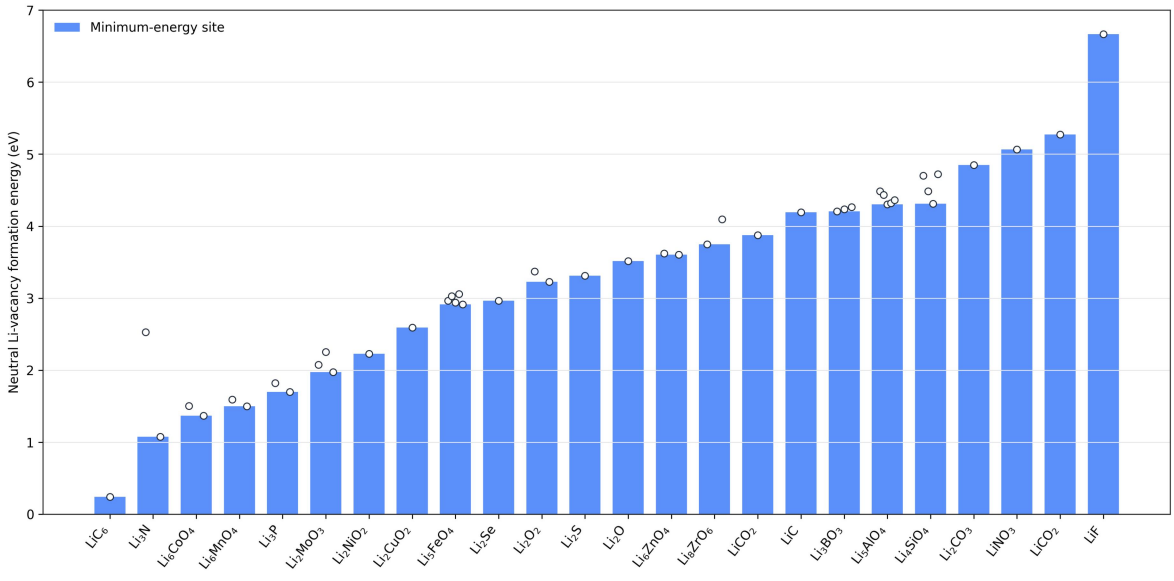


图 1 24 种材料最低中性 Li 空位形成能（同参数 bcc Li 化学势）

最低位点形成能呈现显著材料依赖性。LiC6 的 0.241 eV 远低于其余体系，说明其 Li 脱出在本中性缺陷模型中最容易；Li3N（1.074 eV）、Li6CoO4（1.368 eV）、Li6MnO4（1.499 eV）、Li3P（1.697 eV）和 Li2MoO3（1.971 eV）构成低形成能组。高形成能端依次包括 LiF（6.667 eV）、LiCO2（5.271 eV）、LiNO3（5.067 eV）和 Li2CO3（4.850 eV），体现强离子键/多阴离子骨架对去 Li 过程的较高能量代价。

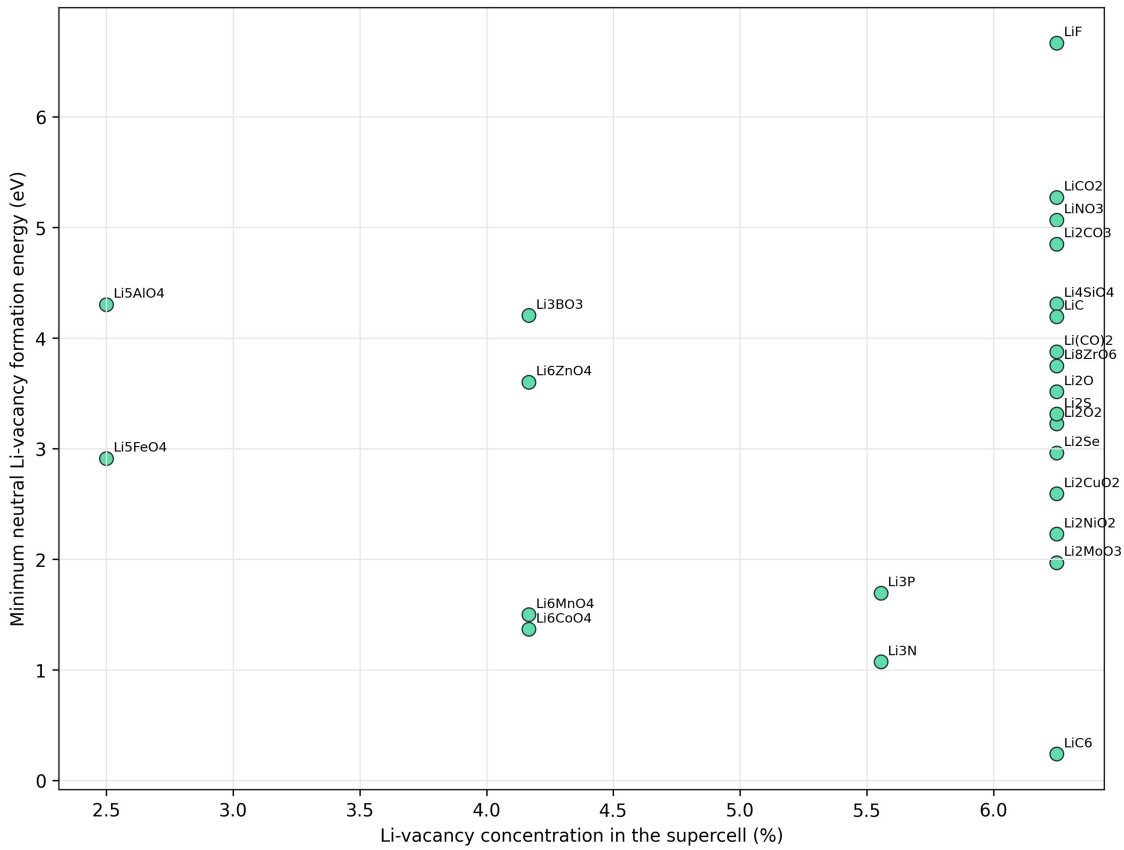


图 2 缺陷浓度与最低 Li 空位形成能；颜色/标签区分材料

形成能与名义 Li 空位浓度的 Pearson 相关系数仅为 r = 0.124（p = 0.564），说明 2.50%–6.25%的浓度差异不是本数据集排序的主要单变量来源。

序号	材料	MP-ID	最低位点	Ef/eV	VLi/%	Fmax/eV Å⁻¹	Li 失电子/e
1	Li5FeO4	mp-19511	5	2.912	2.50	0.0490	0.652
2	Li6CoO4	mp-18925	2	1.368	4.17	0.0442	0.840
3	Li2NiO2	mp-19308	1	2.229	6.25	0.0252	0.635
4	Li2CuO2	mp-4711	1	2.593	6.25	0.0328	0.778
5	Li4SiO4	mp-11737	3	4.310	6.25	0.0385	0.717
6	Li6MnO4	mp-770533	2	1.499	4.17	0.0461	0.816
7	Li6ZnO4	mp-14495	2	3.603	4.17	0.0335	0.902
8	Li5AlO4	mp-15960	3	4.303	2.50	0.0498	0.667
9	Li2MoO3	mp-694855	3	1.971	6.25	0.0497	0.702
10	Li8ZrO6	mp-755225	1	3.748	6.25	0.0342	0.714
11	Li3BO3	mp-27275	1	4.205	4.17	0.0390	0.711
12	LiNO3	mp-8180	1	5.067	6.25	0.0485	0.811
13	Li2O2	mp-841	2	3.227	6.25	0.0281	0.794
14	Li2O	mp-1960	1	3.515	6.25	0.0145	0.873
15	Li2S	mp-1153	1	3.314	6.25	0.0263	0.757
16	Li3N	mp-2251	2	1.074	5.56	0.0483	0.846
17	Li3P	mp-736	2	1.697	5.56	0.0410	0.507
18	Li2Se	mp-2286	1	2.964	6.25	0.0483	0.704
19	LiF	mp-1138	1	6.667	6.25	0.0199	0.905
20	LiC6	mp-1001581	1	0.241	6.25	0.0206	0.920
21	LiC	mp-1378	1	4.195	6.25	0.0384	0.845
22	Li2CO3	mp-3054	1	4.850	6.25	0.0491	0.710
23	LiCO2	mp-1190287	1	5.271	6.25	0.0496	0.789
24	Li(CO)2	mp-546920	1	3.876	6.25	0.0478	0.830

4.1 不等价位点差异

46 个不等价 Li 位点全部保留。位点间能量展宽最大的体系为 Li3N：1.454 eV；Li4SiO4：0.411 eV；Li8ZrO6：0.349 eV；Li2MoO3：0.282 eV。因此，对多 Li 位点材料仅计算一个任意位点可能误判易脱锂位点；本报告以每种材料最低形成能作为主结果，同时在下表保留全部位点数据。

序号	材料	位点	等价原始索引	Ef/eV	Fmax/eV Å⁻¹	最低
1	Li5FeO4	1	0,1,2,3,4,5,6,7	2.964	0.0352	
1	Li5FeO4	2	8,9,10,11,12,13,14,15	3.027	0.0376	
1	Li5FeO4	3	16,17,18,20,21,22,23,39	2.939	0.0391	
1	Li5FeO4	4	19,32,33,34,35,36,37,38	3.057	0.0440	
1	Li5FeO4	5	24,25,26,27,28,29,30,31	2.912	0.0490	是
2	Li6CoO4	1	0,1,2,3	1.503	0.0495	
2	Li6CoO4	2	4,5,6,7,8,9,10,11	1.368	0.0442	是
3	Li2NiO2	1	0,1	2.229	0.0252	是
4	Li2CuO2	1	0,1	2.593	0.0328	是
5	Li4SiO4	1	0,1	4.700	0.0382	
5	Li4SiO4	2	2,3	4.486	0.0465	
5	Li4SiO4	3	4,5	4.310	0.0385	是
5	Li4SiO4	4	6,7	4.722	0.0362	
6	Li6MnO4	1	0,1,2,3	1.591	0.0488	
6	Li6MnO4	2	4,5,6,7,8,9,10,11	1.499	0.0461	是
7	Li6ZnO4	1	0,1,9,10	3.622	0.0407	
7	Li6ZnO4	2	2,3,4,5,6,7,8,11	3.603	0.0335	是
8	Li5AlO4	1	0,1,2,3,4,5,6,7	4.486	0.0444	
8	Li5AlO4	2	8,9,10,11,12,13,14,15	4.433	0.0415	
8	Li5AlO4	3	16,17,18,20,21,22,23,39	4.303	0.0498	是
8	Li5AlO4	4	19,32,33,34,35,36,37,38	4.319	0.0489	
8	Li5AlO4	5	24,25,26,27,28,29,30,31	4.364	0.0450	
9	Li2MoO3	1	0,1	2.075	0.0344	
9	Li2MoO3	2	2	2.253	0.0492	
9	Li2MoO3	3	3	1.971	0.0497	是
10	Li8ZrO6	1	0,1,2,5,6,7	3.748	0.0342	是

序号	材料	位点	等价原始索引	Ef/eV	Fmax/eV Å⁻ ¹	最低
10	Li8ZrO6	2	3,4	4.098	0.0369	
11	Li3BO3	1	0,1,2,3	4.205	0.0390	是
11	Li3BO3	2	4,5,6,7	4.237	0.0421	
11	Li3BO3	3	8,9,10,11	4.267	0.0470	
12	LiNO3	1	0,1	5.067	0.0485	是
13	Li2O2	1	0,1	3.372	0.0372	
13	Li2O2	2	2,3	3.227	0.0281	是
14	Li2O	1	0,1	3.515	0.0145	是
15	Li2S	1	0,1	3.314	0.0263	是
16	Li3N	1	0	2.527	0.0449	
16	Li3N	2	1,2	1.074	0.0483	是
17	Li3P	1	0,1	1.821	0.0179	
17	Li3P	2	2,3,4,5	1.697	0.0410	是
18	Li2Se	1	0,1	2.964	0.0483	是
19	LiF	1	0	6.667	0.0199	是
20	LiC6	1	0	0.241	0.0206	是
21	LiC	1	0,1	4.195	0.0384	是
22	Li2CO3	1	0,1,2,3	4.850	0.0491	是
23	LiCO2	1	0,1,2,3	5.271	0.0496	是
24	Li(CO)2	1	0,1	3.876	0.0478	是

5. Bader 电荷分析

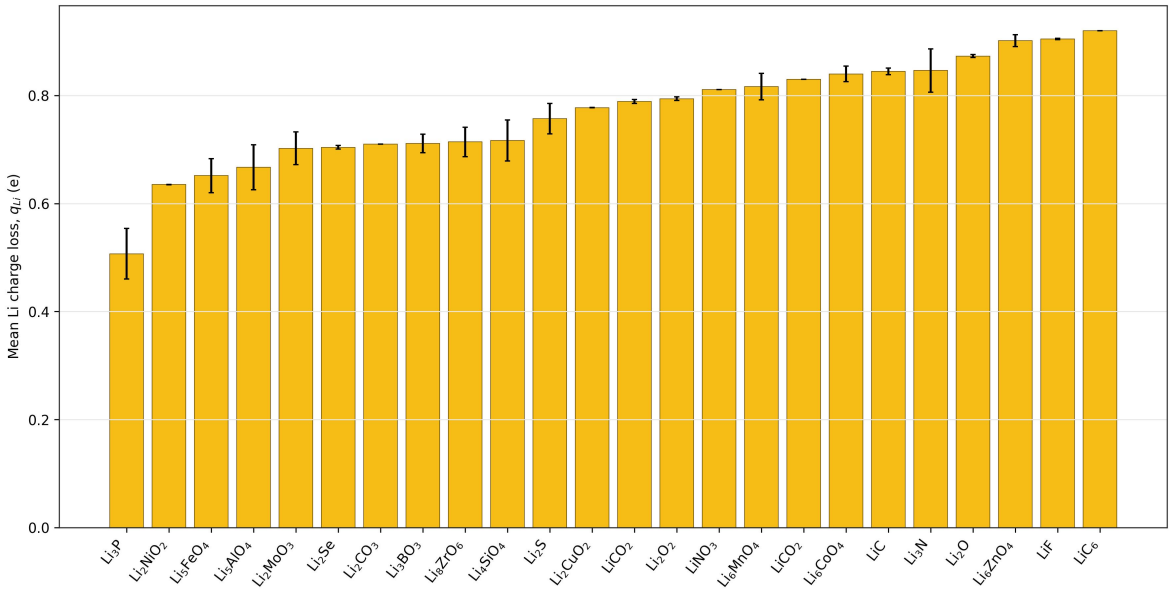


图 3 24 种 bulk 材料中 Li 的平均 Bader 净失电子量及原子间离散度

24 种材料中的 Li 均表现为净失电子，平均值介于+0.507 e（Li3P）和+0.920 e（LiC6）之间。Bader 值描述的是基于电荷密度零通量面分区得到的电子重排，不能直接等同于形式氧化态；但在同一计算协议下，可用于比较不同化学环境中 Li—骨架电荷转移强弱。

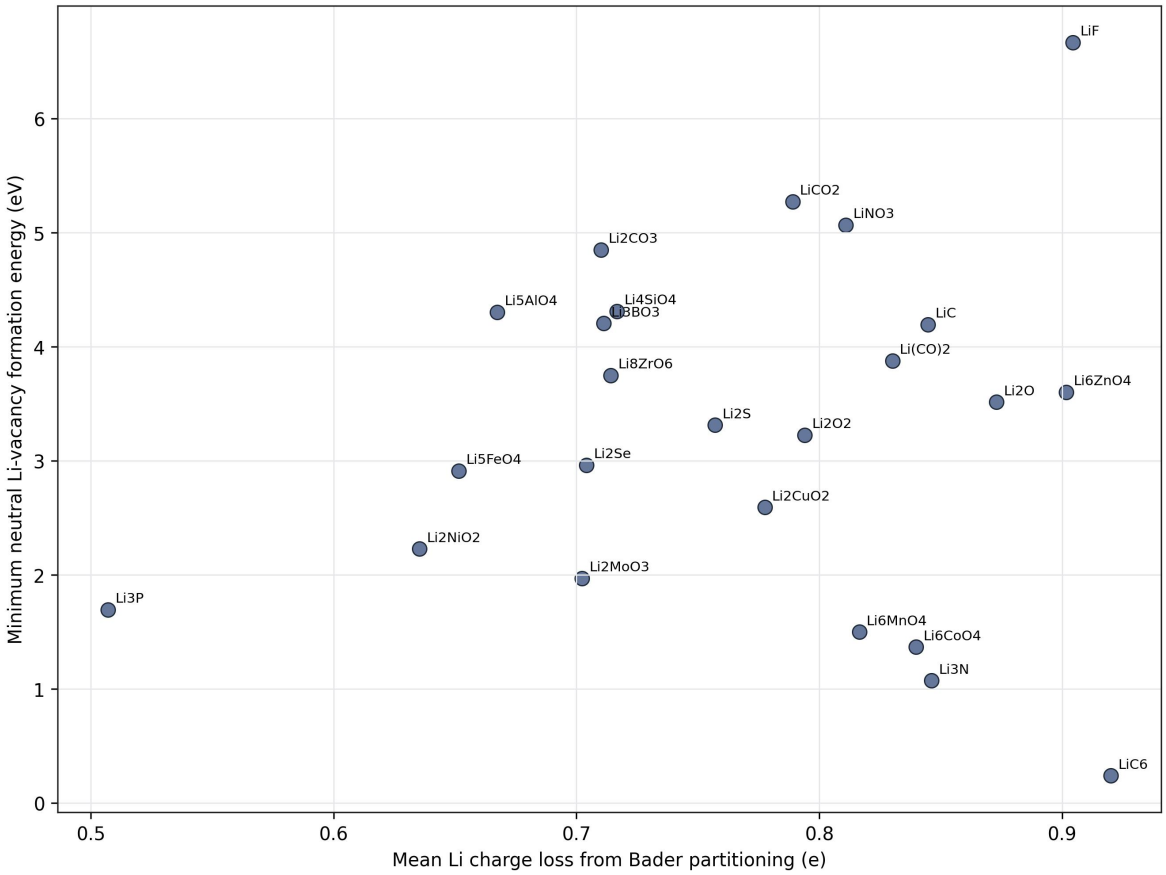


图 4 Li 平均 Bader 净失电子量与最低 Li 空位形成能的关系

Li 平均失电子量与最低形成能的 Pearson 相关系数为 r = 0.077（p = 0.720），未观察到显著线性关系。例如 LiC6 同时具有最高 Li 失电子量和最低形成能，而 LiF 则在高形成能区，这说明去 Li 能量还取决于宿主骨架重构、局域配位、空位弛豫以及电子补偿方式。完整逐原子 Bader 值和各元素统计见随报告交付的 CSV。

6. ELF 分布与局域化特征

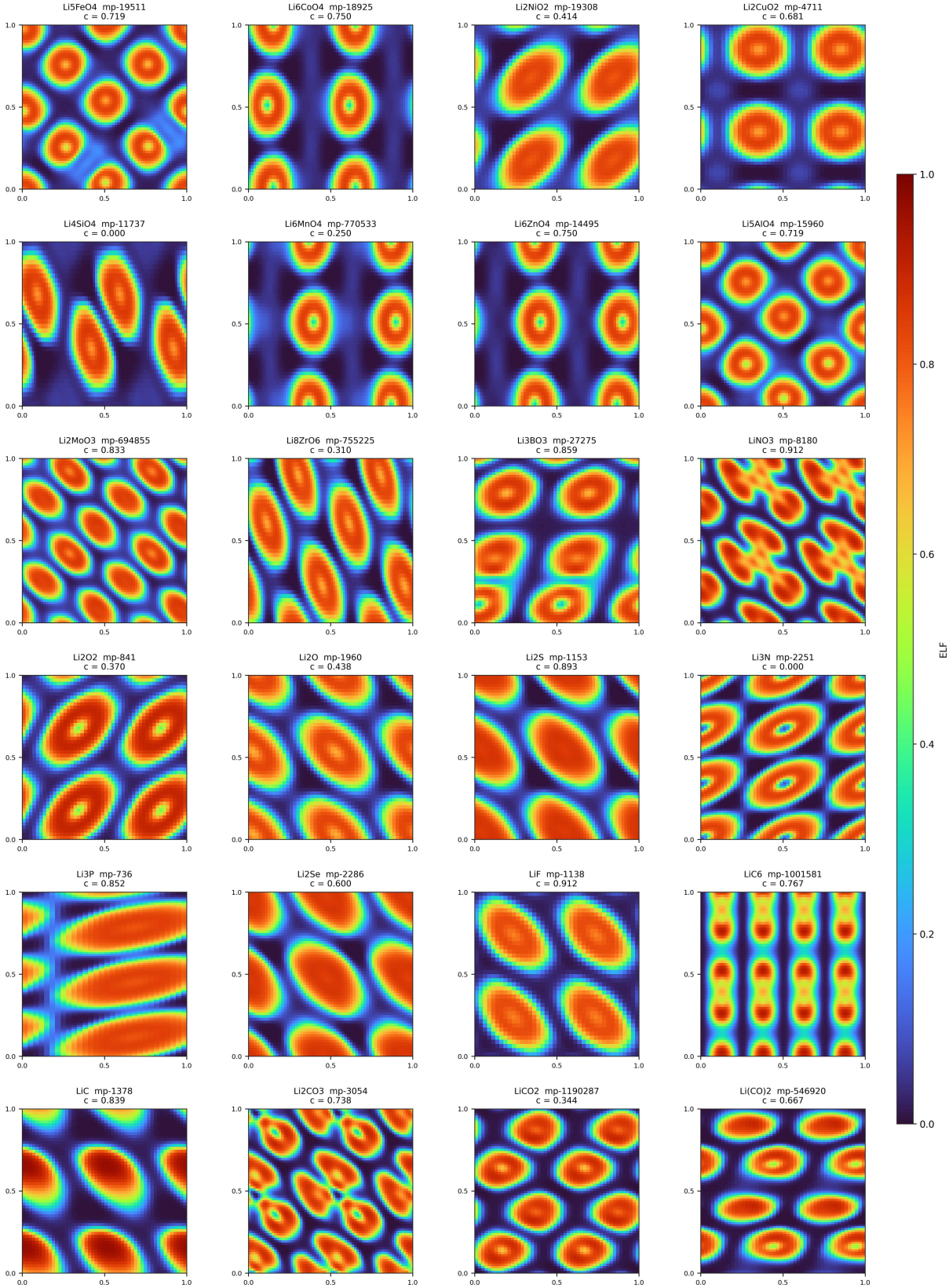


图 5 24 种 bulk 材料的代表性 ELF 切片（统一 0–1 色标）

ELF 图显示不同阴离子骨架和键合类型具有明显差异：高 ELF 区域通常集中于强局域化电子对或共价键附近，低 ELF 区域对应较离域或低电子局域程度的空间。完整三维统计中，全局平均 ELF 从 Li2CuO2 的 0.2483 到 Li3P 的 0.3954。由于各结构的代表切片位置按网格方差自动选择，跨材料比较应优先结合统一色标、三维统计量和化学环境，而不应仅凭某一切片的局部颜色判断。

最低形成能与全局平均 ELF 的 Pearson 相关系数为 r = -0.177（p = 0.409），未显示强线性关系；与 ELF≥0.8 体积分数的 Spearman 相关为 ρ = 0.453（p = 0.026）。后者提示高局域化体积分数可能与较高去 Li 代价同向变化，但样本数仅 24，且材料类别差异显著，因此只能作为后续同系材料筛选的探索性指标。

序号	材料	平均 ELF	ELF≥0.8 体积分数/%	最低 Ef/eV
1	Li5FeO4	0.2743	6.52	2.912
2	Li6CoO4	0.2751	6.29	1.368
3	Li2NiO2	0.2691	6.54	2.229
4	Li2CuO2	0.2483	5.62	2.593
5	Li4SiO4	0.3036	7.91	4.310
6	Li6MnO4	0.2769	5.96	1.499
7	Li6ZnO4	0.2798	6.38	3.603
8	Li5AlO4	0.3010	7.66	4.303
9	Li2MoO3	0.3070	6.95	1.971
10	Li8ZrO6	0.3071	7.90	3.748
11	Li3BO3	0.3024	7.68	4.205
12	LiNO3	0.3103	9.62	5.067
13	Li2O2	0.3037	9.72	3.227
14	Li2O	0.3006	7.26	3.515
15	Li2S	0.3662	15.46	3.314
16	Li3N	0.3132	5.75	1.074
17	Li3P	0.3954	13.84	1.697
18	Li2Se	0.3724	15.69	2.964
19	LiF	0.2758	6.53	6.667
20	LiC6	0.3488	5.11	0.241
21	LiC	0.3563	10.94	4.195
22	Li2CO3	0.2963	8.15	4.850
23	LiCO2	0.2952	7.82	5.271
24	Li(CO)2	0.2893	7.09	3.876

7. 综合讨论

（1）形成能排序首先反映宿主材料的整体化学键合与去 Li 后的电子/结构响应，而不是单一 Li 原子的静态电荷。Bader 和 ELF 提供互补信息：Bader 量化原子盆中的电荷转移，ELF 揭示空间局域化；两者均应与缺陷弛豫能量共同解读。

（2）LiC6、Li3N、Li6CoO4、Li6MnO4、Li3P 和 Li2MoO3 在本计算条件下具有较低 Li 空位形成能，适合作为容易发生脱 Li 或 Li 非化学计量的候选；LiF、LiCO2、LiNO3 和 Li2CO3 的高形成能表明生成中性 Li 空位代价较高。该结论针对中性空位和 Li 金属化学势，不代表带电缺陷在任意费米能级、氧化电位或温度下的全部热力学行为。

（3）位点展宽在 Li3N 等材料中不可忽略，说明结构对称性降低后的局域环境会影响去 Li 难易。后续若用于电化学电压、实际非化学计量或动力学研究，建议对低形成能候选进一步进行超胞尺寸收敛、带电缺陷/费米能级分析、相稳定性以及 Li 迁移 NEB 计算。

8. 结论

• 完成 24 种 bulk、46 个不等价 Li 空位和 bcc Li 参考相的 GPU-VASP 计算；全部任务成功并满足规定的 ENCUT 与力收敛阈值。

• 24 种材料最低中性 Li 空位形成能为 0.241–6.667 eV；LiC6 最低，LiF 最高，全部为正值。

• Li 平均 Bader 净失电子量为+0.507–+0.920 e，但其与形成能的线性相关很弱，不能单独作为跨化学体系预测器。

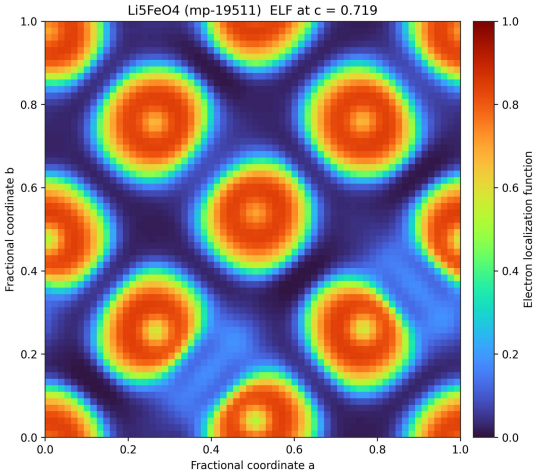
• ELF 三维统计与形成能同样不存在强的平均值线性关系；高 ELF 体积分数的中等 Spearman 相关仅作为探索性信号。

• 报告保留全部 46 个位点结果、24 套 Bader 与 ELF 原始/派生数据、71 张 CrystalStudio 结构图及完整参数证据，可支持逐项复核与二次制图。

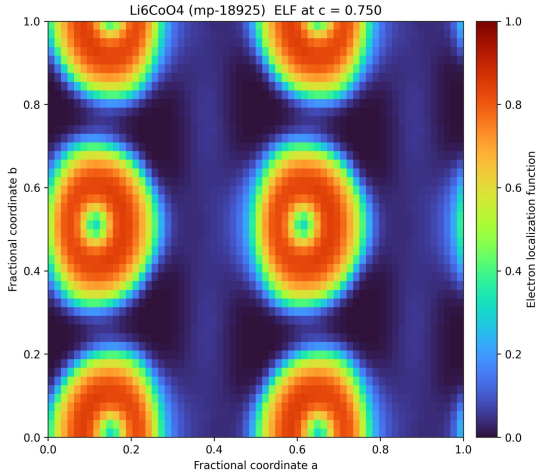
附录 A 24 种材料 ELF 分布图集

下列 24 张图均由对应 bulk 静态 ELFCAR 生成，采用统一 0–1 色标；切片位置为沿 c 方向网格标准差最大的代表平面。

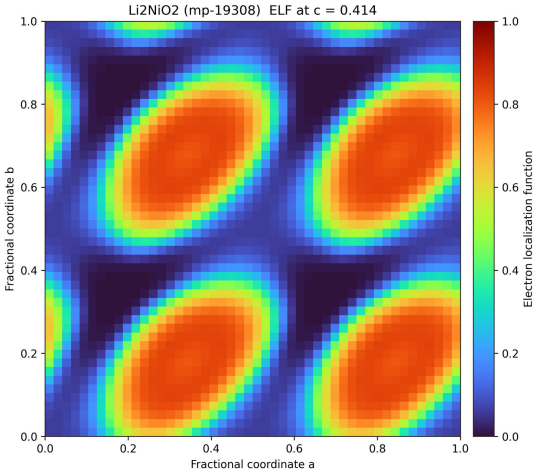
ELF 图集 A-1



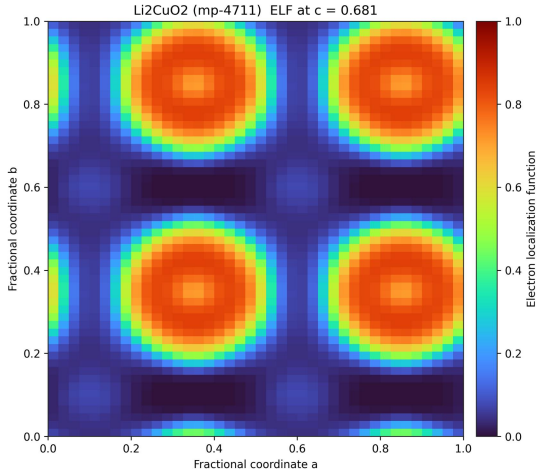
01_Li5FeO4_mp-19511



02_Li6CoO4_mp-18925

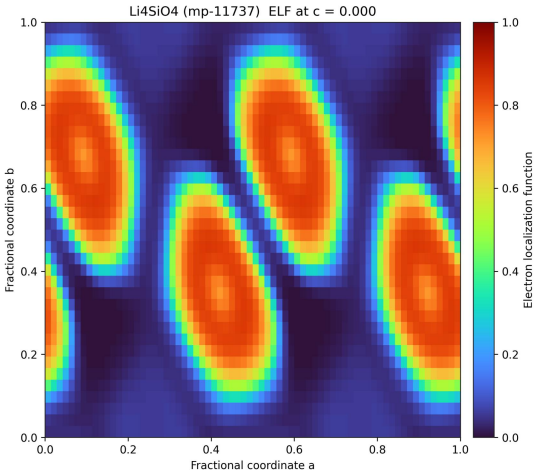


03_Li2NiO2_mp-19308

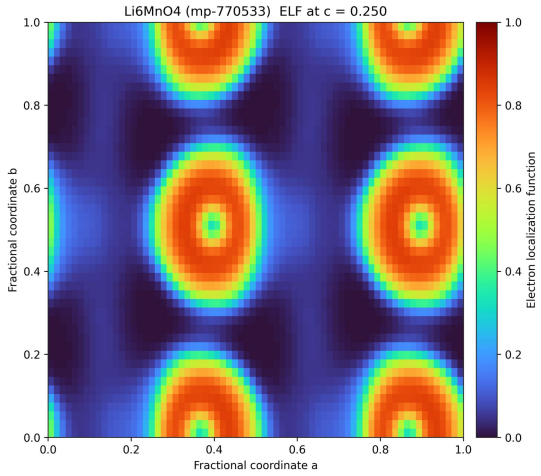


04_Li2CuO2_mp-4711

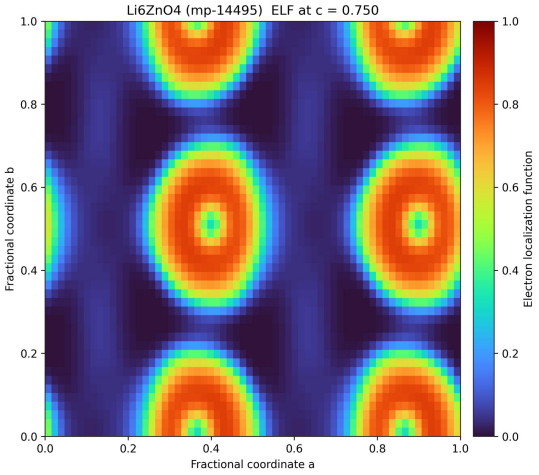
ELF 图集 A-2



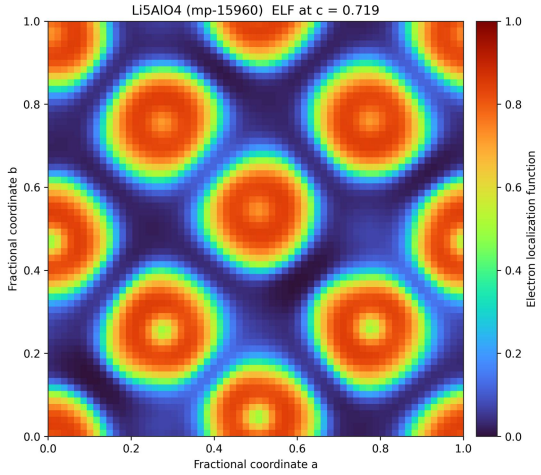
05_Li4SiO4_mp-11737



06_Li6MnO4_mp-770533

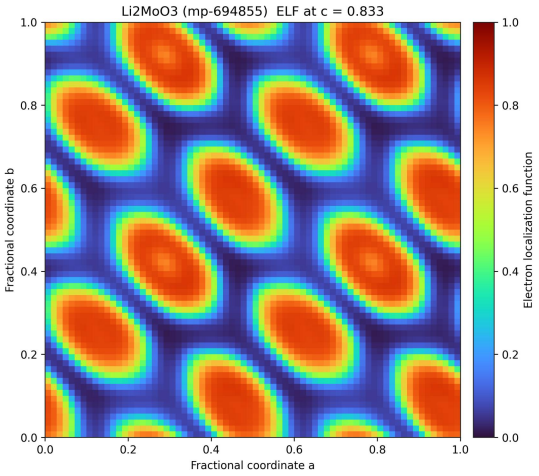


07_Li6ZnO4_mp-14495

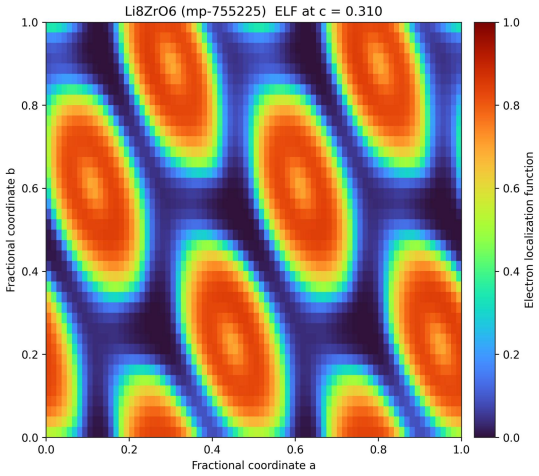


08_Li5AlO4_mp-15960

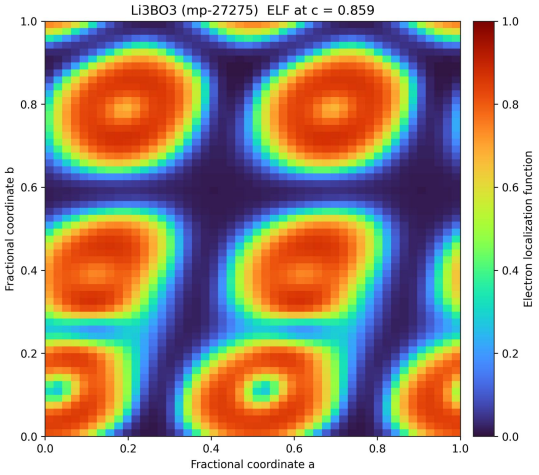
ELF 图集 A-3



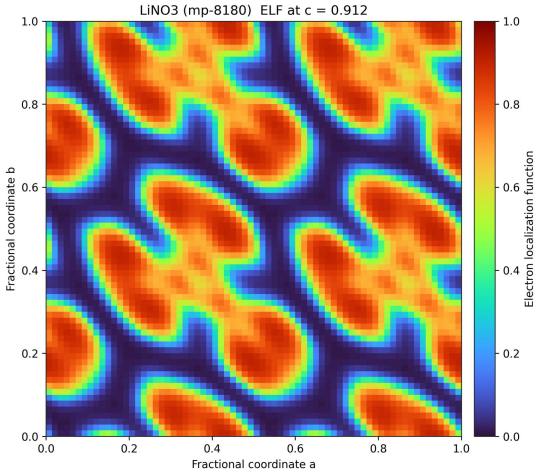
09_Li2MoO3_mp-694855



10_Li8ZrO6_mp-755225

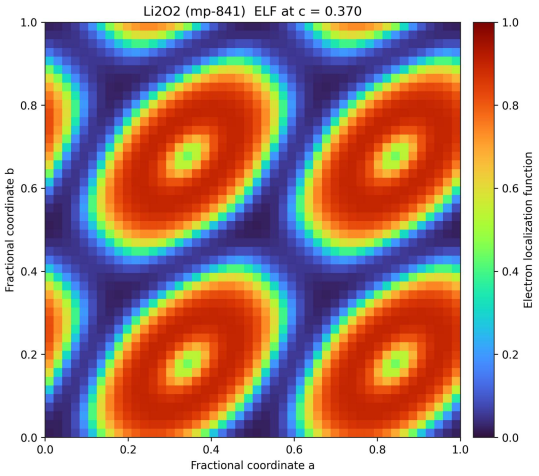


11_Li3BO3_mp-27275

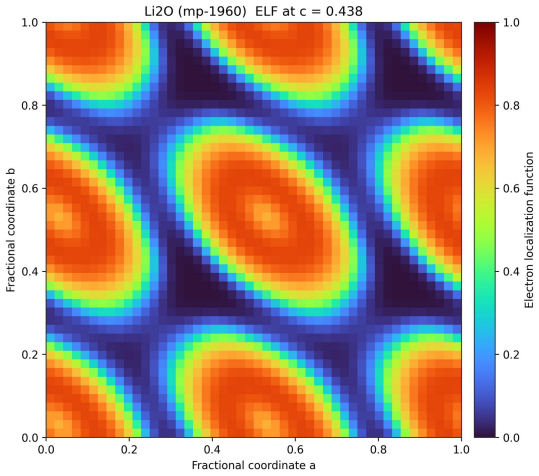


12_LiNO3_mp-8180

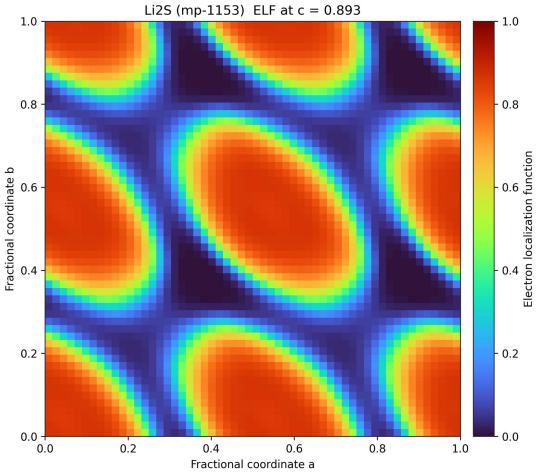
ELF 图集 A-4



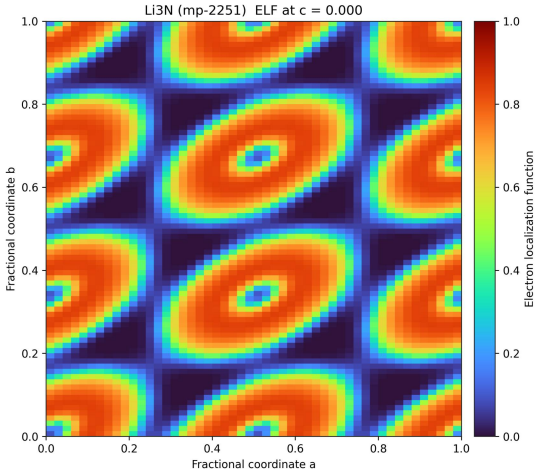
13_Li2O2_mp-841



14_Li2O_mp-1960

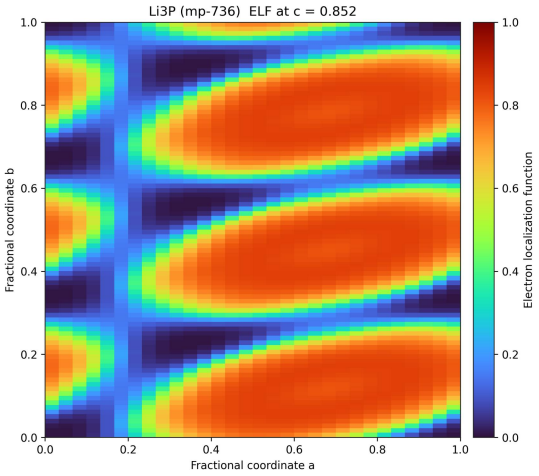


15_Li2S_mp-1153

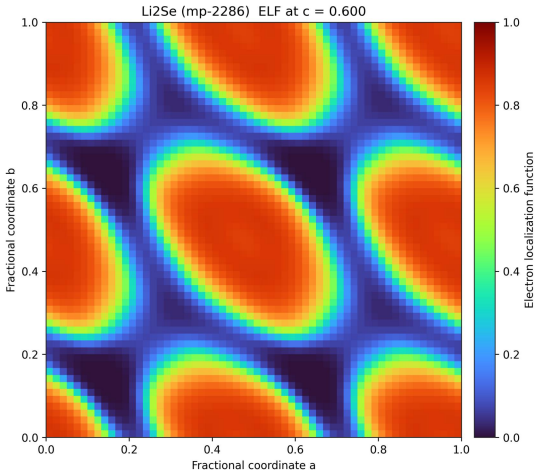


16_Li3N_mp-2251

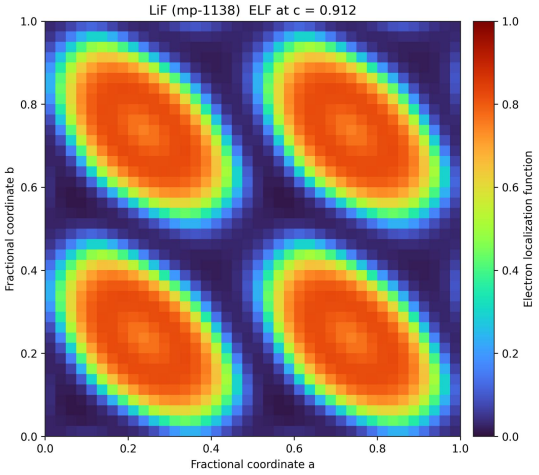
ELF 图集 A-5



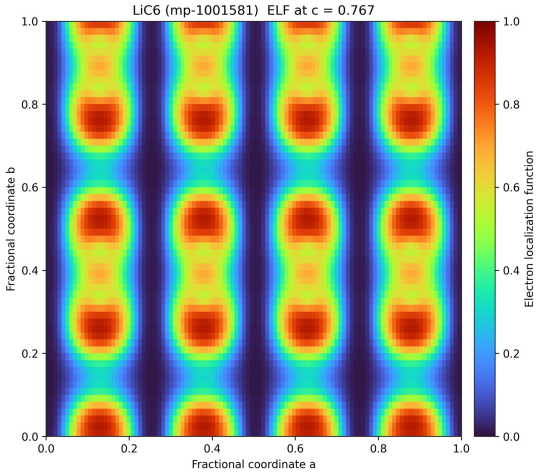
17_Li3P_mp-736



18_Li2Se_mp-2286

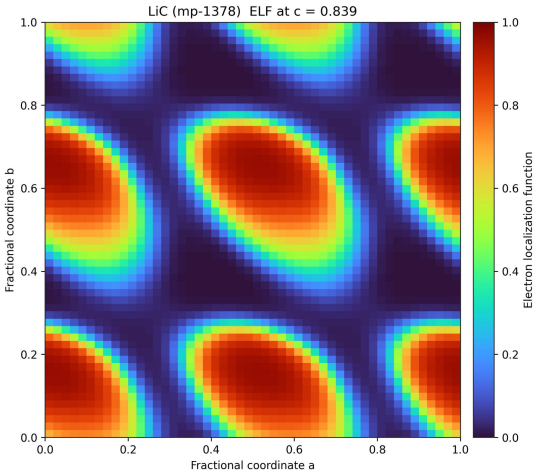


19_LiF_mp-1138

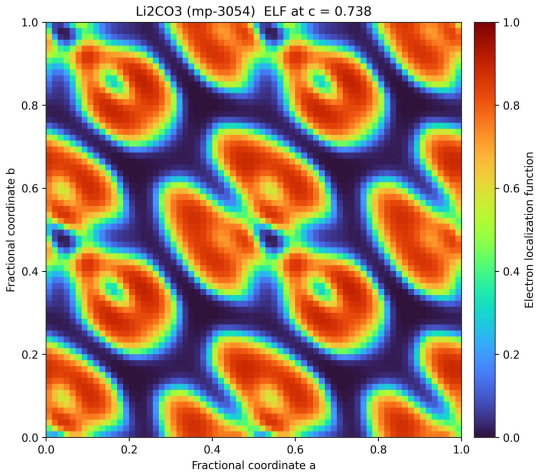


20_LiC6_mp-1001581

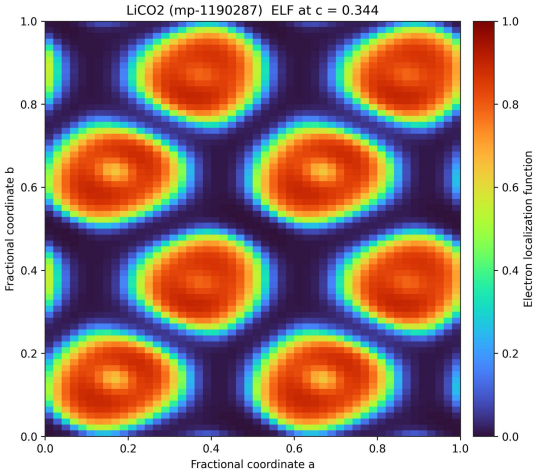
ELF 图集 A-6



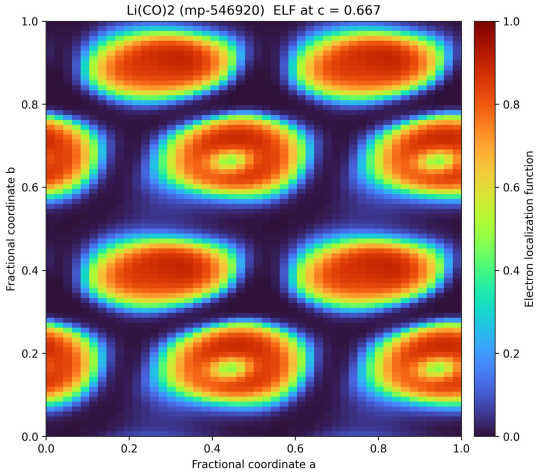
21_LiC_mp-1378



22_Li2CO3_mp-3054



23_LiCO2_mp-1190287

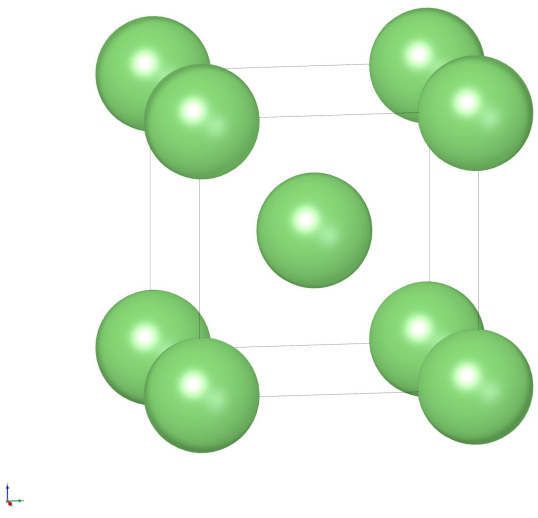


24_Li(CO)2_mp-546920

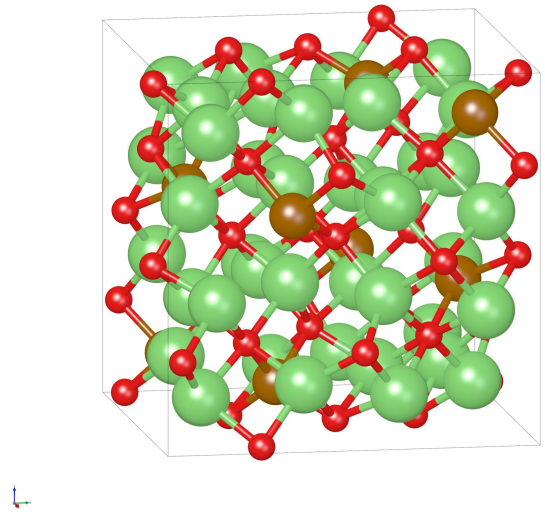
附录 B 完整结构模型图集

本附录覆盖 bcc Li 参考相、24 个 bulk 结构和 46 个不等价 Li 空位结构，共 71 张。全部图由 CrystalStudio 0.12.30 后台（隐藏 Electron）模式渲染；a、b、c 方向显示范围均为−0.01–1.01，晶胞线框为黑色、线宽 1.0，白色背景，统一等轴视角并自动适配结构。

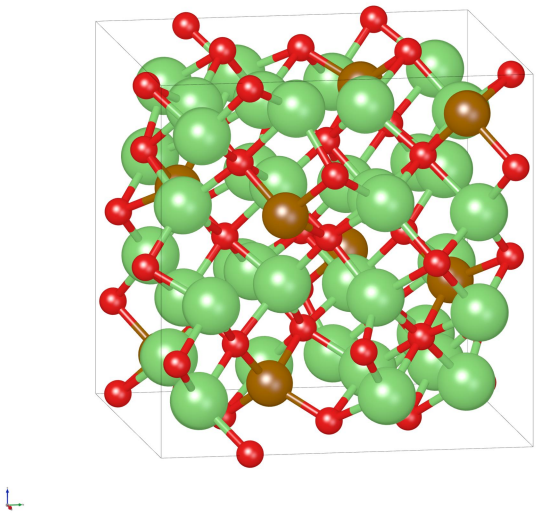
结构图集 B-1



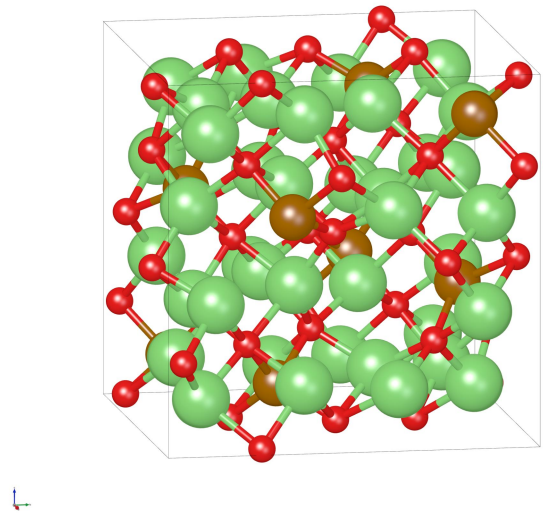
00_Li_bcc_reference



01_Li5FeO4_mp-19511 bulk

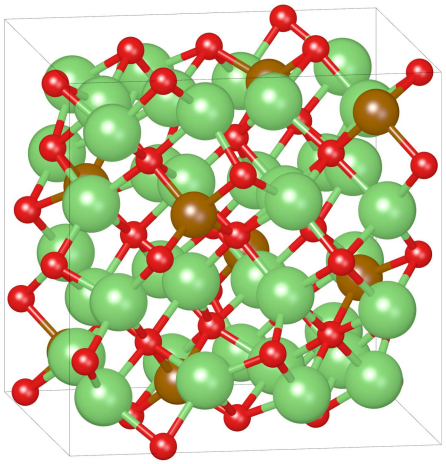


01_Li5FeO4_mp-19511 VLi site01

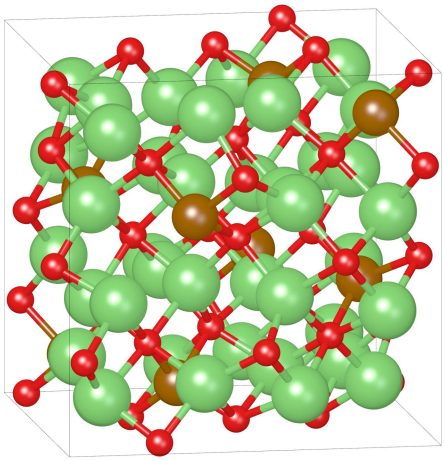


01_Li5FeO4_mp-19511 VLi site02

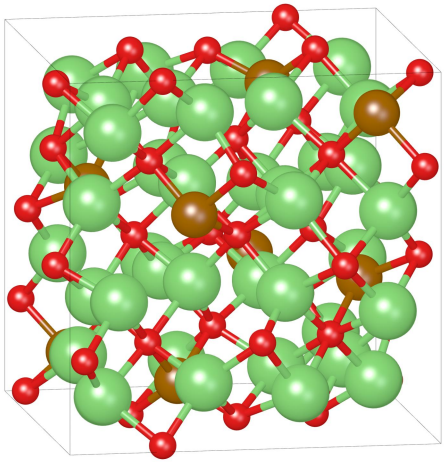
结构图集 B-2



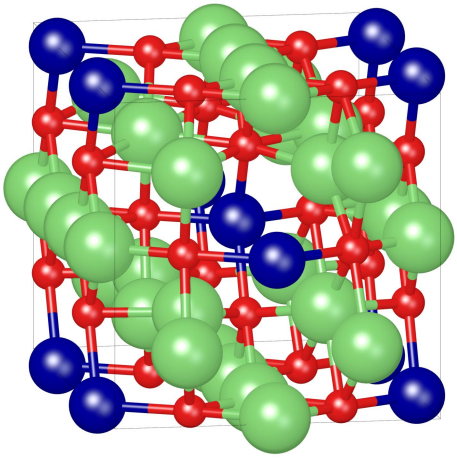
01_Li5FeO4_mp-19511 VLi site03



01_Li5FeO4_mp-19511 VLi site04

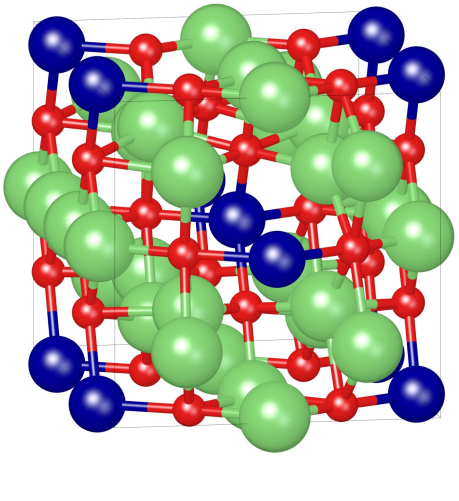


01_Li5FeO4_mp-19511 VLi site05

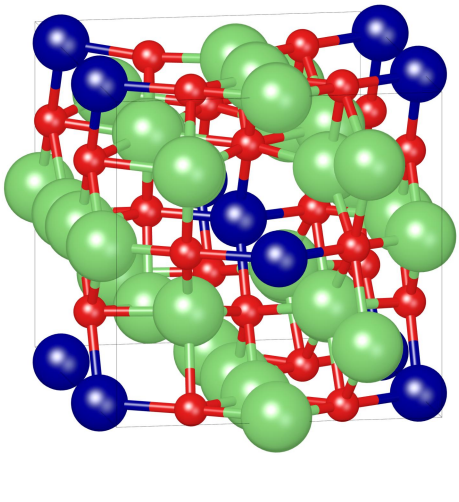


02_Li6CoO4_mp-18925 bulk

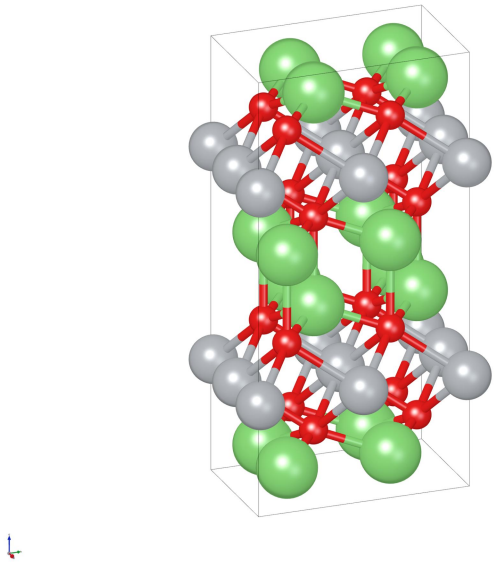
结构图集 B-3



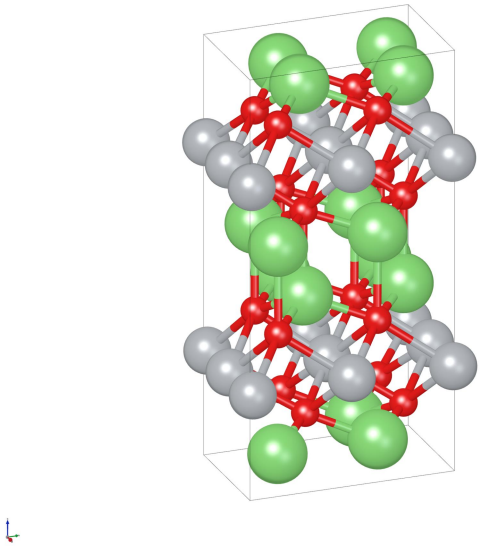
02_Li6CoO4_mp-18925 VLi site01



02_Li6CoO4_mp-18925 VLi site02

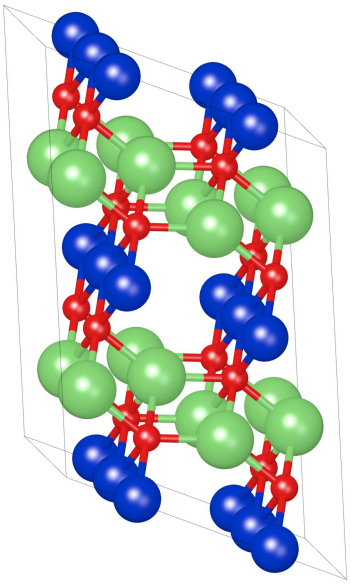


03_Li2NiO2_mp-19308 bulk

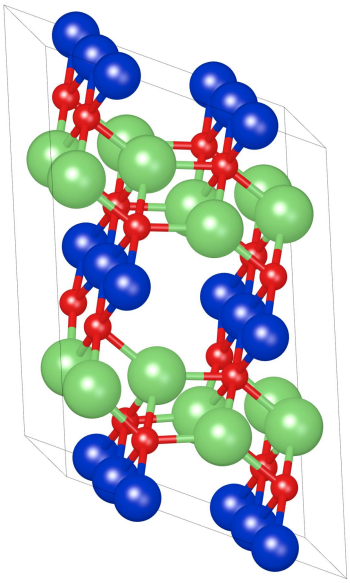


03_Li2NiO2_mp-19308 VLi site01

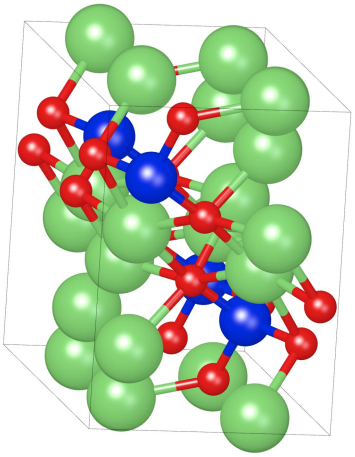
结构图集 B-4



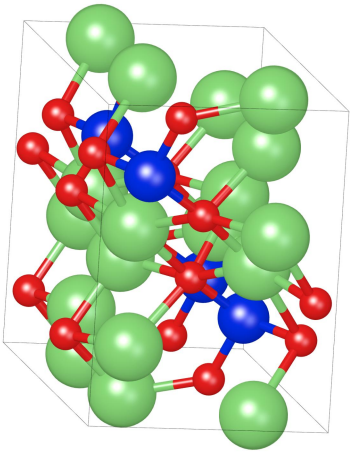
04_Li2CuO2_mp-4711 bulk



04_Li2CuO2_mp-4711 VLi site01

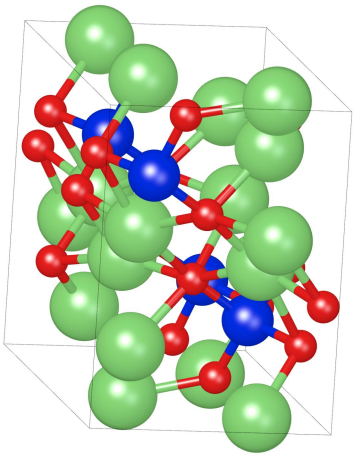


05_Li4SiO4_mp-11737 bulk

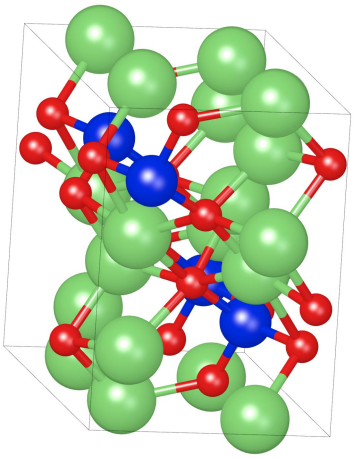


05_Li4SiO4_mp-11737 VLi site01

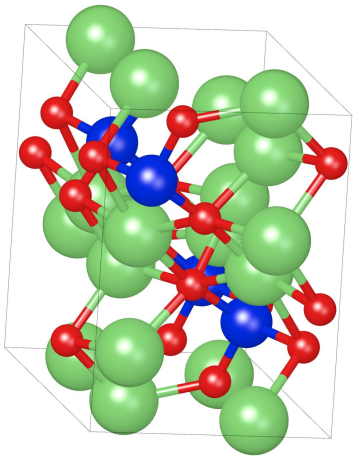
结构图集 B-5



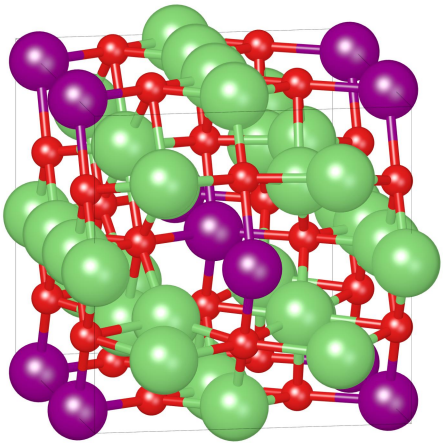
05_Li4SiO4_mp-11737 VLi site02



05_Li4SiO4_mp-11737 VLi site03

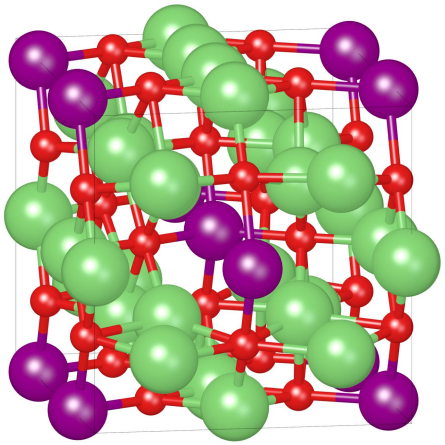


05_Li4SiO4_mp-11737 VLi site04

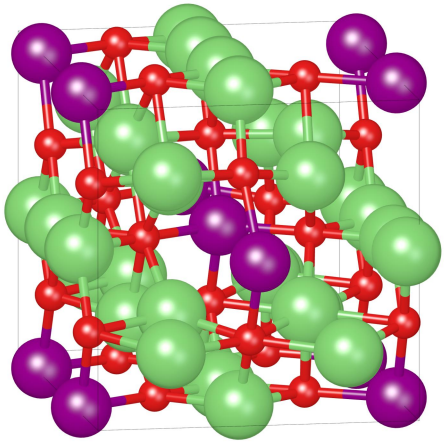


06_Li6MnO4_mp-770533 bulk

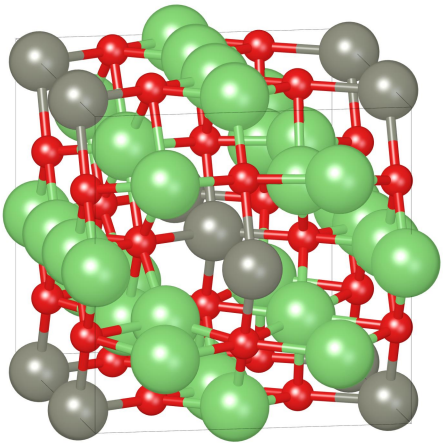
结构图集 B-6



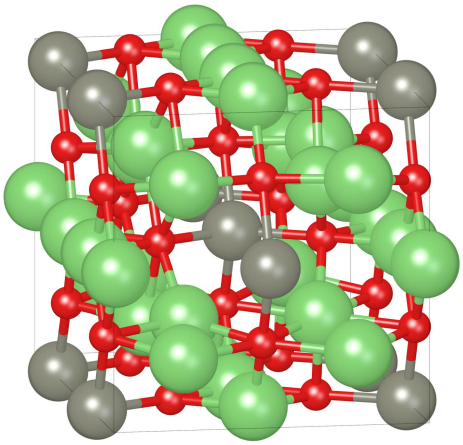
06_Li6MnO4_mp-770533 VLi site01



06_Li6MnO4_mp-770533 VLi site02

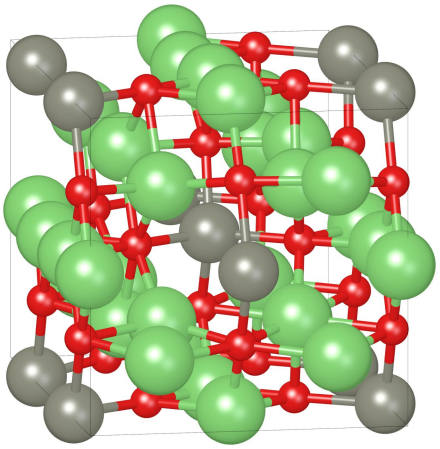


07_Li6ZnO4_mp-14495 bulk

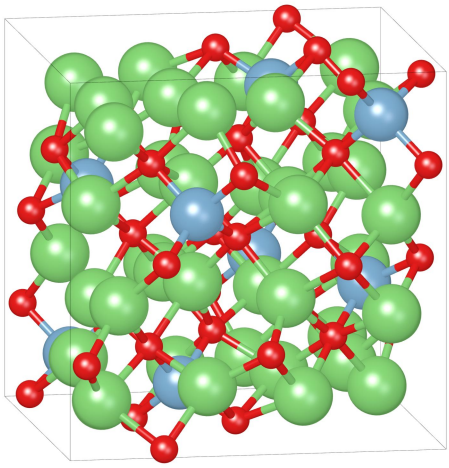


07_Li6ZnO4_mp-14495 VLi site01

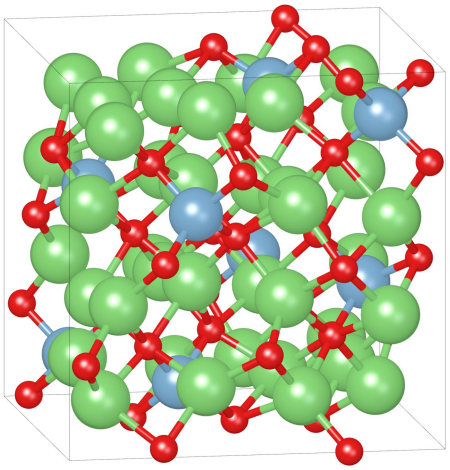
结构图集 B-7



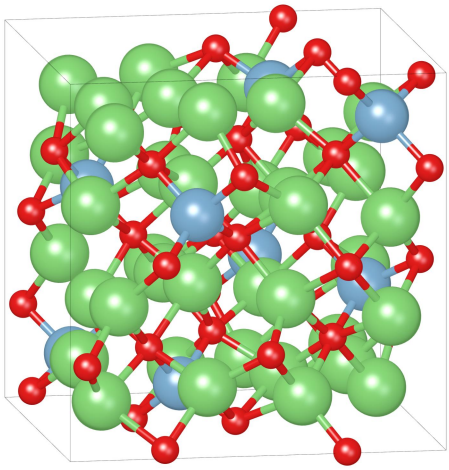
07_Li6ZnO4_mp-14495 VLi site02



08_Li5AlO4_mp-15960 bulk

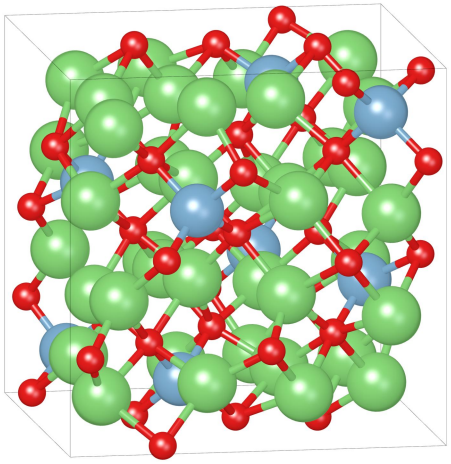


08_Li5AlO4_mp-15960 VLi site01

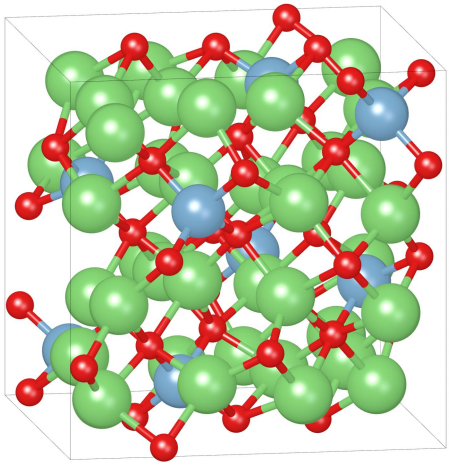


08_Li5AlO4_mp-15960 VLi site02

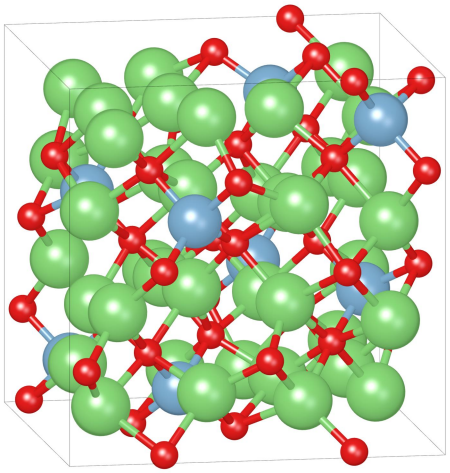
结构图集 B-8



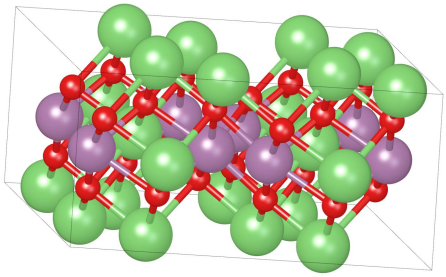
08_Li5AlO4_mp-15960 VLi site03



08_Li5AlO4_mp-15960 VLi site04

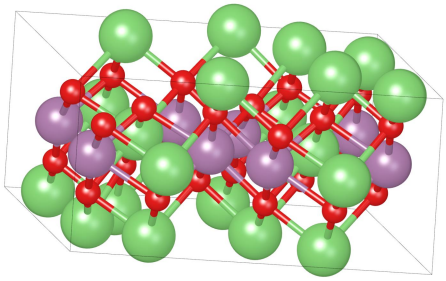


08_Li5AlO4_mp-15960 VLi site05

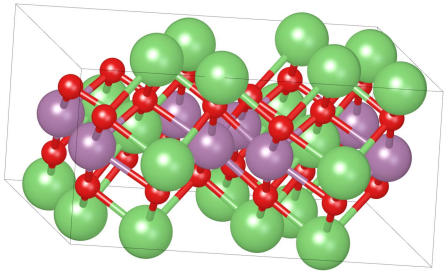


09_Li2MoO3_mp-694855 bulk

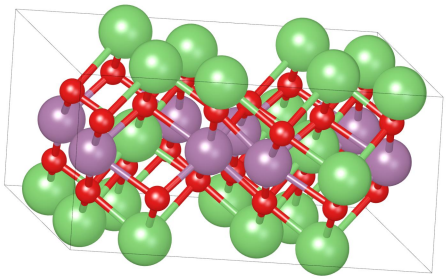
结构图集 B-9



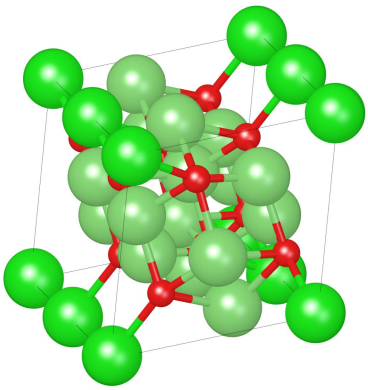
09_Li2MoO3_mp-694855 VLi site01



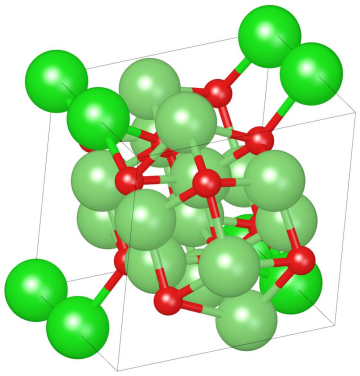
09_Li2MoO3_mp-694855 VLi site02



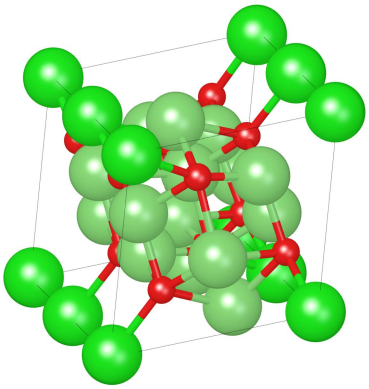
09_Li2MoO3_mp-694855 VLi site03



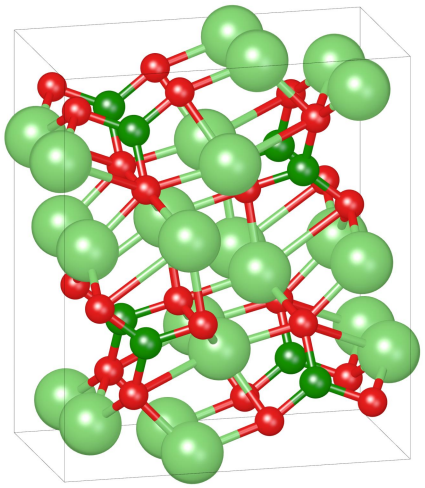
10_Li8ZrO6_mp-755225 bulk



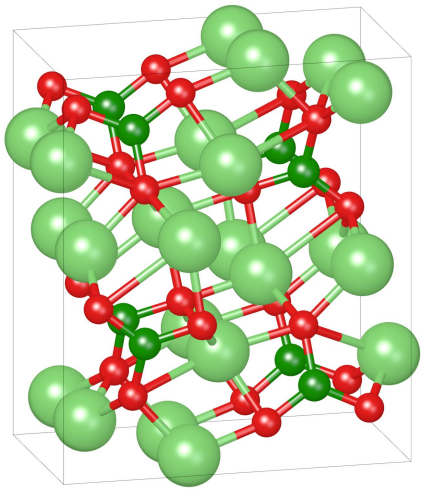
10_Li8ZrO6_mp-755225 VLi site01



10_Li8ZrO6_mp-755225 VLi site02

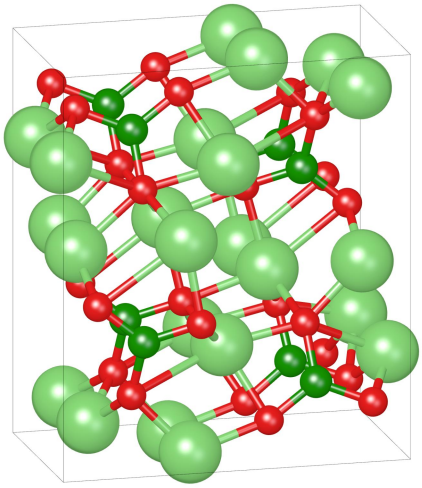


11_Li3BO3_mp-27275 bulk

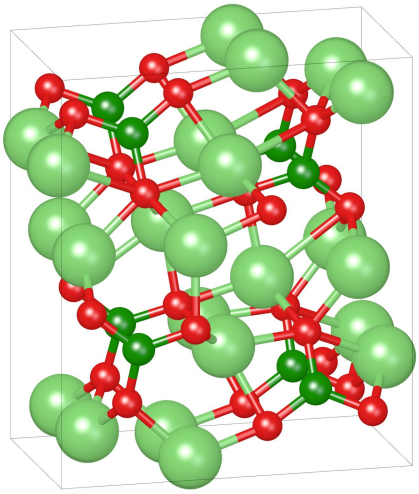


11_Li3BO3_mp-27275 VLi site01

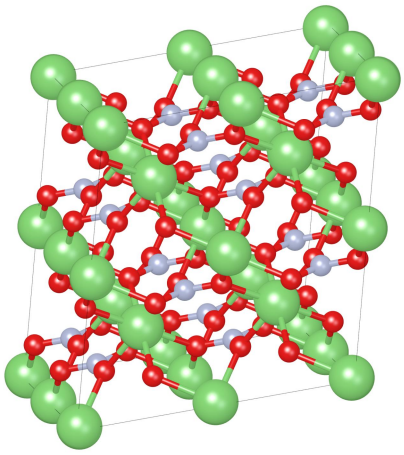
结构图集 B-11



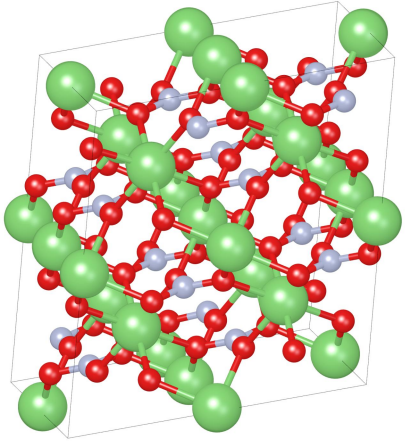
11_Li3BO3_mp-27275 VLi site02



11_Li3BO3_mp-27275 VLi site03

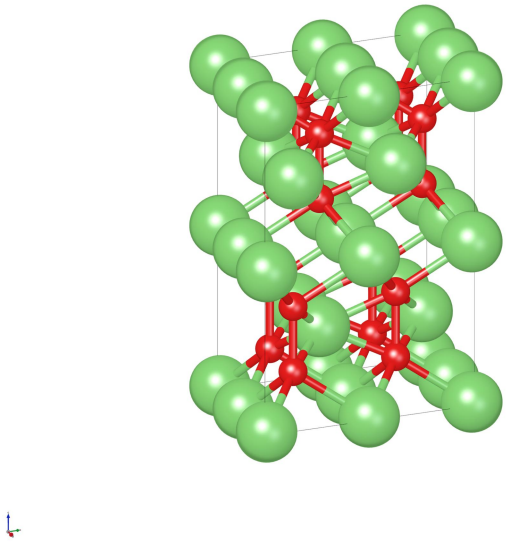


12_LiNO3_mp-8180 bulk

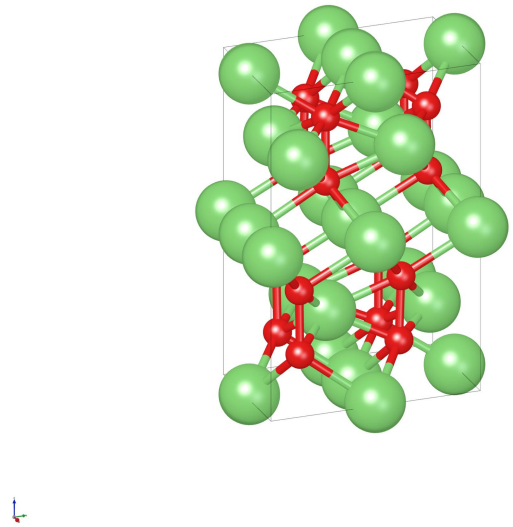


12_LiNO3_mp-8180 VLi site01

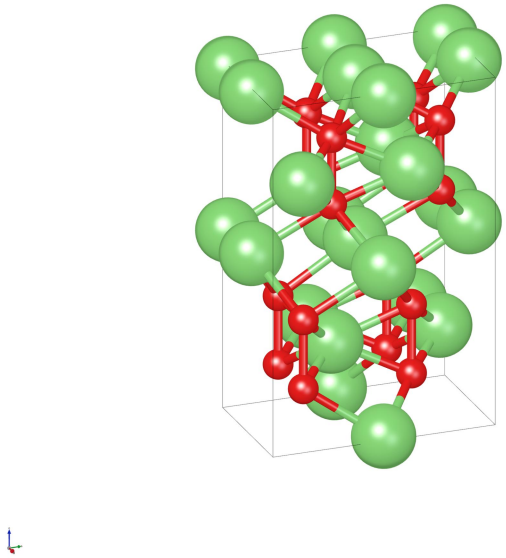
结构图集 B-12



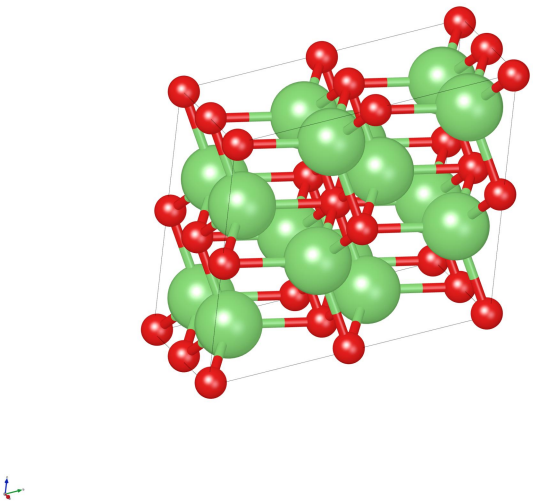
13_Li2O2_mp-841 bulk



13_Li2O2_mp-841 VLi site01

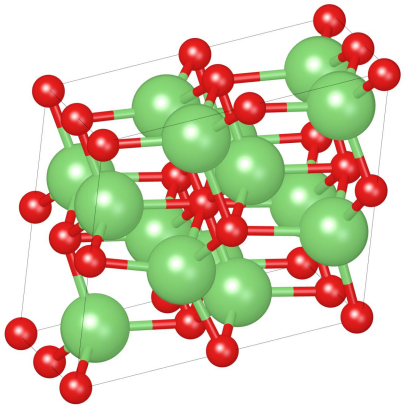


13_Li2O2_mp-841 VLi site02

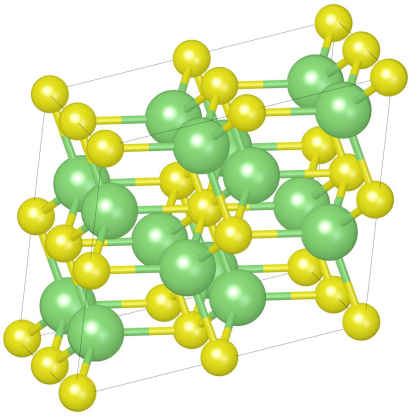


14_Li2O_mp-1960 bulk

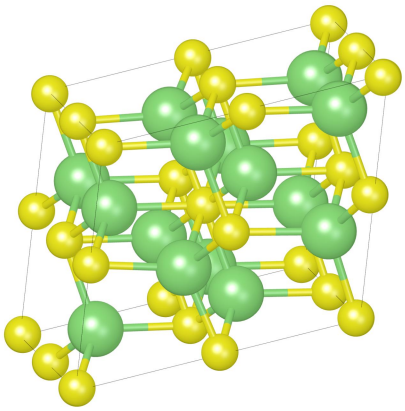
结构图集 B-13



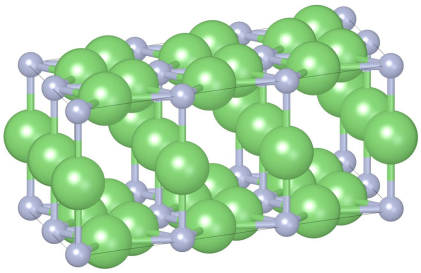
14_Li2O_mp-1960 VLi site01



15_Li2S_mp-1153 bulk

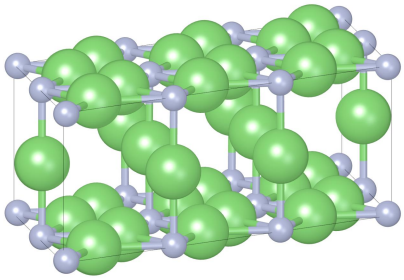


15_Li2S_mp-1153 VLi site01

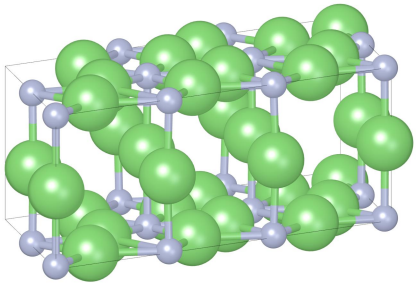


16_Li3N_mp-2251 bulk

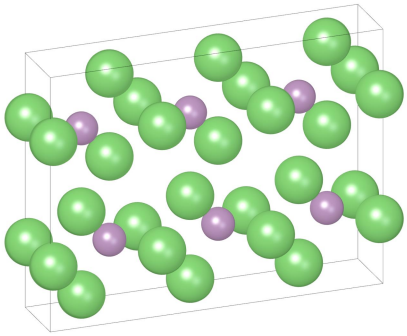
结构图集 B-14



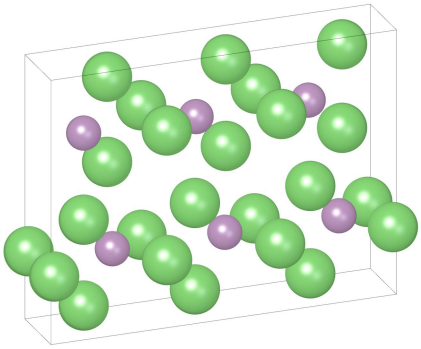
16_Li3N_mp-2251 VLi site01



16_Li3N_mp-2251 VLi site02

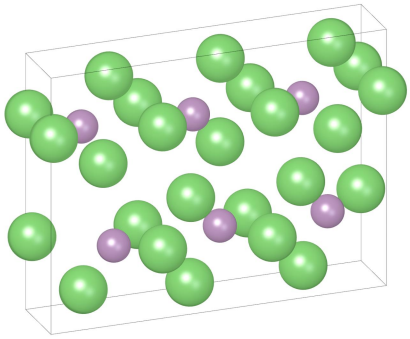


17_Li3P_mp-736 bulk

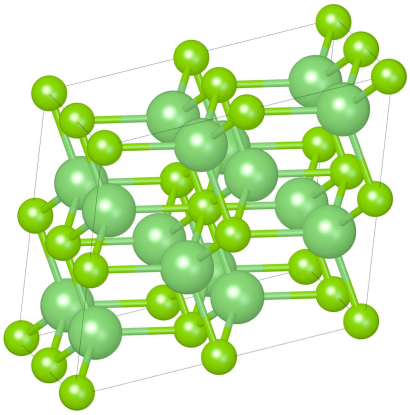


17_Li3P_mp-736 VLi site01

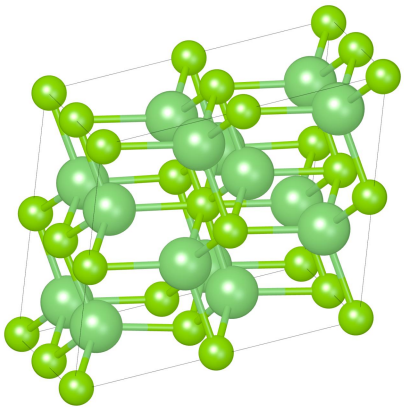
结构图集 B-15



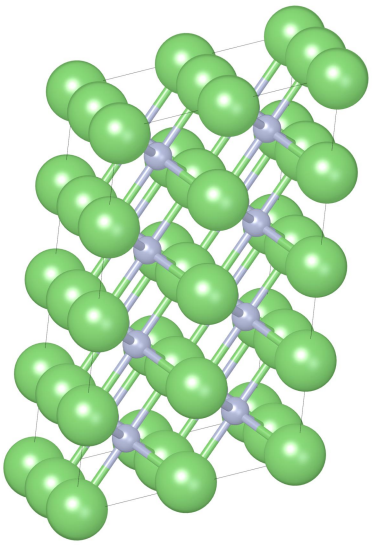
17_Li3P_mp-736 VLi site02



18_Li2Se_mp-2286 bulk

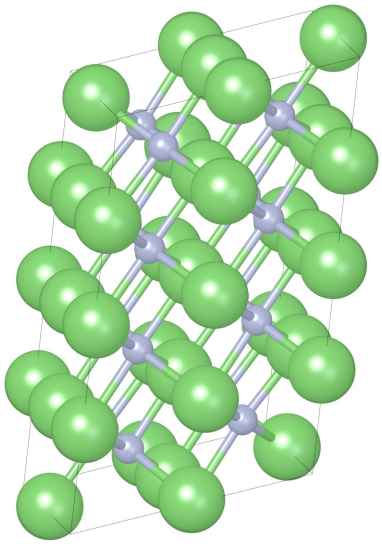


18_Li2Se_mp-2286 VLi site01

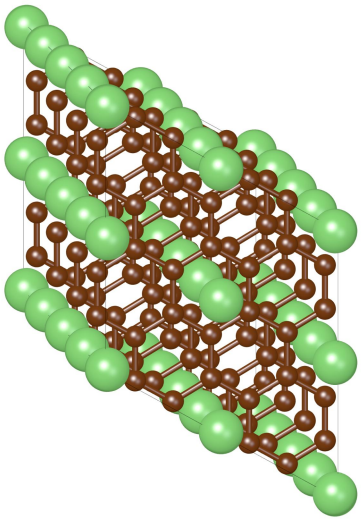


19_LiF_mp-1138 bulk

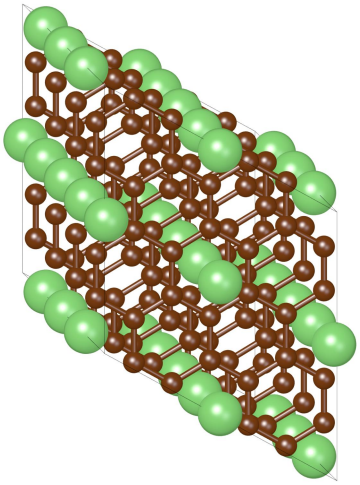
结构图集 B-16



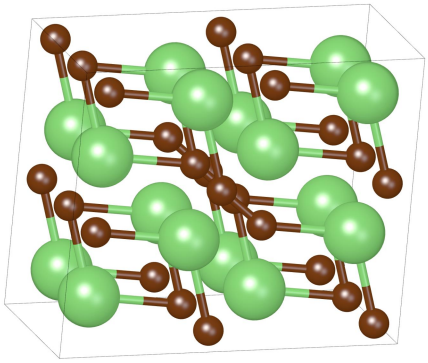
19_LiF_mp-1138 VLi site01



20_LiC6_mp-1001581 bulk

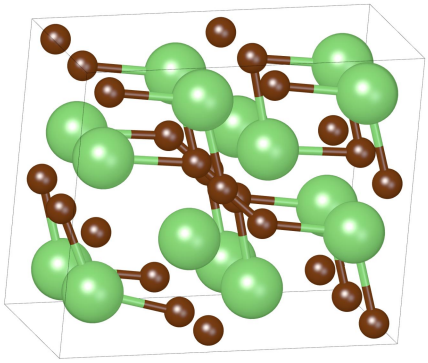


20_LiC6_mp-1001581 VLi site01

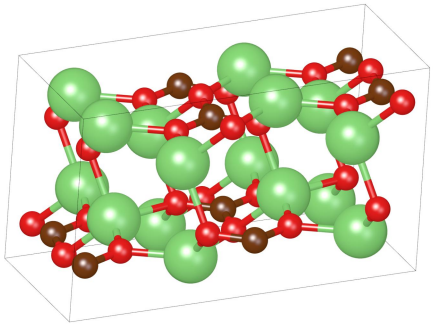


21_LiC_mp-1378 bulk

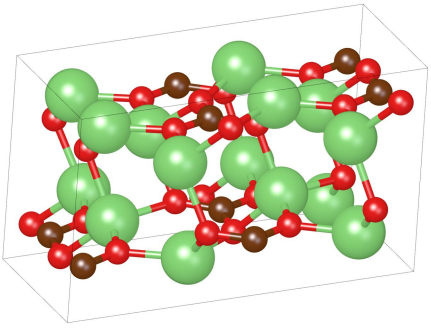
结构图集 B-17



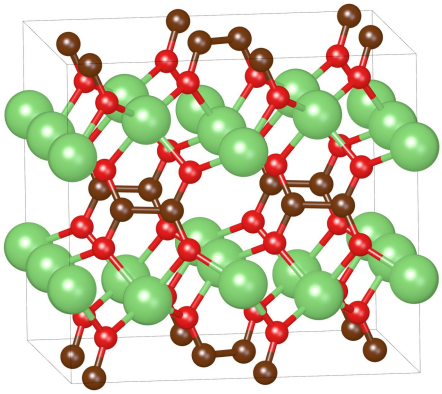
21_LiC_mp-1378 VLi site01



22_Li2CO3_mp-3054 bulk

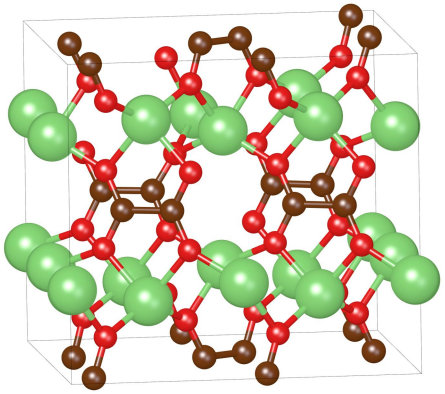


22_Li2CO3_mp-3054 VLi site01

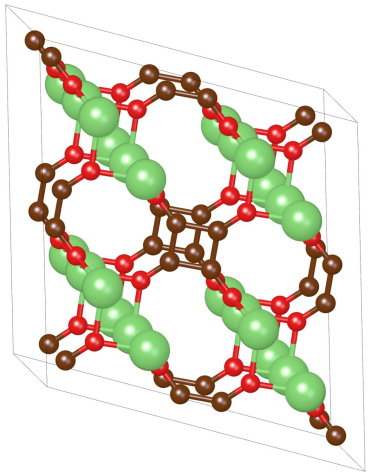


23_LiCO2_mp-1190287 bulk

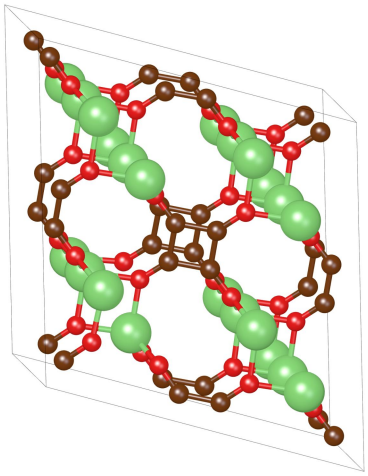
结构图集 B-18



23_LiCO2_mp-1190287 VLi site01



24_Li(CO)2_mp-546920 bulk



24_Li(CO)2_mp-546920 VLi site01

Computational details

Spin-polarized density-functional theory calculations were performed with VASP 6.5.1[1] using the PBE generalized-gradient approximation[2] and projector-augmented-wave potentials.[3,4] The plane-wave cutoff was 450 eV. Electronic and ionic convergence criteria were 1×10^{-5} eV and 0.05 eV Å^{-1}, respectively. Calculations used PREC = Accurate, ISMEAR = 0, SIGMA = 0.05 eV, LREAL = .FALSE., LASPH = .TRUE., and spin polarization. Γ-centered k-point meshes were selected for each supercell; the exact meshes are listed in the structure table. Structural relaxations used IBRION = 2, except for two continued mp-18925 vacancy relaxations that used IBRION = 1 as explicitly requested. Static calculations used IBRION = −1 and NSW = 0.

For pristine bulk structures, CHGCAR, AECCAR0, AECCAR2, and ELFCAR were written. Bader partitioning used the valence charge density with AECCAR0 + AECCAR2 as the all-electron reference.[5] The reported charge loss is q = Zval − QBader, so q > 0 denotes electron loss. Electron localization functions were interpreted following Becke and Edgecombe.[6] ELF maps use the c-normal grid plane with the largest standard deviation in each material and a common 0–1 color scale. Full three-dimensional ELFCAR grids were used for global means and threshold-volume fractions.

Neutral Li-vacancy formation energies were calculated as Ef = Edefect − Ebulk + μLi, where μLi = −1.885471885 eV atom^{-1} was obtained from bcc Li using the same electronic-structure settings. For each material, all symmetry-inequivalent Li sites were retained and the lowest formation energy is used as the primary comparison. The selected minimum-cost supercells yield nominal Li-vacancy concentrations of 2.50–6.25%; therefore, the comparison is internally consistent in method but not a strict infinite-dilution limit.

References

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附录 C 原子图例

适用于附录 B 的全部结构模型图。下列色标采用 CrystalStudio 0.12.30 的 VESTA 元素基础配色；球体高光和阴影会使局部明暗有所变化。

- Li 锂 #87E074
- B 硼 #20A20F
- C 碳 #81492A
- N 氮 #B0BAE6
- O 氧 #FF0300
- F 氟 #B0BAE6
- Al 铝 #81B3D6
- Si 硅 #1B3BFA
- P 磷 #C19CC3
- S 硫 #FFFA00
- Mn 锰 #A9099E
- Fe 铁 #B57200
- Co 钴 #0000AF
- Ni 镍 #B8BCC0
- Cu 铜 #2247DD
- Zn 锌 #8F9082
- Se 硒 #9AEF10
- Zr 锆 #00FF00
- Mo 钼 #B486B0

说明：N 与 F 在该配色方案中使用相同基础色，需结合材料化学式区分。图例色标大小仅用于识别元素，不表示原子半径、价态或电荷大小；Li 空位表示移除 Li 原子后的位置，不对应新增原子色标。